



Graduation Project Proposal and Report Handbook

Darp: AI-Based Early Risk Detection in Crowded and High-Temperature Environments

Computer Science and Artificial Intelligence

College of Computing

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Abstract

Large-scale mass gatherings such as the Hajj season present significant health and safety challenges due to extreme crowd density and elevated temperatures, increasing the risks of heat exhaustion and fainting. Current solutions often address crowd management and health monitoring separately, creating a gap in integrated real-time risk prevention.

This project proposes a lightweight AI-based prototype for early risk detection in crowded and high-temperature environments. The system employs a Convolutional Neural Network (CNN), trained on the ShanghaiTech dataset, to classify crowd density with a target accuracy of at least 85%. Crowd density is integrated with normalized temperature values and user-reported symptoms through a rule-based engine to compute a weighted baseline risk score.

The system generates location-based alerts to support medical teams in prioritizing rapid intervention in high-risk zones. It also provides preventive notifications for pilgrims within the same area without offering medical diagnosis or storing personal data.

The expected outcome is a fully functional web-based prototype that enables spatial risk monitoring and supports timely decision-making while preserving privacy.

Keywords: Hajj Safety, Crowd Density Estimation, Heat Risk Assessment, Early Warning System.



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Chapter 1

INTRODUCTION AND MOTIVATION

1.1 Problem Context and Application Domain

This project falls within the field of Artificial Intelligence, with a focus on Computer Vision and its applications in risk management within crowded environments. The system targets large-scale mass gatherings such as the Hajj season, where high crowd density intersects with extreme environmental conditions.

The core problem lies in the increased likelihood of sudden health risks, such as heat exhaustion and fainting, under these conditions. Detecting early warning signs and responding promptly becomes challenging due to the dynamic and highly congested nature of such environments.

Although existing solutions provide crowd monitoring or health tracking, they operate independently, resulting in a lack of a comprehensive, real-time understanding of the overall risk level.

1.2 Motivation and Significance

The motivation behind this project stems from the need to overcome the limitations of current systems that rely on partial analysis of the problem.

Effective risk assessment in crowded environments requires the integration of multiple interrelated factors, including crowd density, environmental conditions, and basic health indicators. Relying on a single factor leads to incomplete evaluation and delayed response.

This project is significant as it introduces an integrated approach that combines these factors within a unified framework, enhancing early risk detection and enabling more proactive and informed decision-making in high-risk environments.

1.3 High-Level Project Vision

The project aims to develop a lightweight intelligent system capable of performing real-time risk assessment by integrating multiple data sources.

The system utilizes Computer Vision techniques to estimate crowd density, alongside environmental temperature data and simple user-reported health indicators. By combining these inputs, the system generates an accurate risk level assessment for each monitored area.

The system is designed to support authorities and medical teams through continuous monitoring and real-time alerts, while also allowing users to interact by reporting simple symptoms and receiving preventive notifications.



1.4 Expected Academic or Practical Impact

The expected outcome of this project is the development of an intelligent system capable of dynamically classifying risk levels into clear categories, enabling faster and more effective decision-making.

Practically, the system enhances response efficiency by identifying high-risk areas and generating early alerts, which supports better resource allocation and reduces the likelihood of incidents. Academically, the project demonstrates the value of integrating multiple data sources within a unified AI-based framework and highlights the impact of such integration on improving real-time risk assessment in complex environments.



Chapter 2

BACKGROUND AND RELATED WORK

The studies included in this review were selected to ensure relevance, quality, and reliability. Priority was given to studies published from 2020 onwards to reflect recent advancements in Artificial Intelligence. However, three earlier references were included because they provide important foundations for this project. Reference [14] contributed a foundational crowd counting model, reference [22] provides a methodology for constructing weighted risk models, and reference [20] presents fundamental concepts related to heat stress and environmental risk assessment. The selected studies cover crowd analysis, heat-related risk assessment, and AI-based healthcare monitoring systems. Only peer-reviewed journal articles and conference papers from reputable publishers were considered, with preference given to studies that provide experimental results and practical applications. Modern techniques such as deep learning, CNNs, transformer-based models, and IoT technologies were also included. Furthermore, the selected studies represent different perspectives, including crowd monitoring, healthcare systems, and environmental risk assessment. This diversity supports the development of a comprehensive and practical AI-based solution for real-world crowded environments.

2.1 Relevant Concepts and Technologies

Recent advancements in Artificial Intelligence (AI) have significantly enhanced the ability to analyze complex and dynamic environments such as mass gatherings. In particular, Computer Vision and Deep Learning, especially Convolutional Neural Networks (CNNs), have become the core technologies for crowd analysis and scene understanding.

Crowd analysis has evolved from traditional image processing techniques to deep learning-based models capable of estimating density and detecting congestion in real time. Recent surveys highlight that CNN-based methods significantly improve accuracy in crowd density estimation, especially in dense and occluded environments [1, 2]. Transformer-based architectures have also been introduced to improve feature extraction in highly congested scenes [3].

In parallel, health monitoring systems have gained attention with the integration of Artificial Intelligence and Internet of Things (IoT) technologies. These systems allow continuous monitoring of health-related indicators and support early detection of potential risks. Recent studies show that AI-based healthcare systems can process real-time data and improve decision-making in emergency situations [4,5].

Another critical concept is heat-related risk modeling, which is especially important in extreme environments such as Hajj. Research indicates that prolonged exposure to high temperatures significantly increases the risk of heat stroke and heat exhaustion, particularly in crowded environments where mobility is limited [6]. The World Health Organization also emphasizes the importance of preventive monitoring systems during mass gatherings to reduce heat-related health risks [7].



2.2 Existing Systems and Solutions

Existing systems in the literature address crowd monitoring, digital health, and environmental risk analysis; however, they differ in scope, performance, and level of integration.

2.2.1 *Digital Health Systems*

Digital health systems such as [8, 9] focus on symptom-checking and user interaction, improving accessibility and response time in non-emergency situations. However, these systems rely heavily on user input and lack awareness of environmental conditions such as crowd density and temperature. More advanced healthcare monitoring systems in [4, 5] integrate Artificial Intelligence (AI) and the Internet of Things (IoT), enabling continuous monitoring and improved decision-making. Despite these advantages, they depend on wearable devices and still do not incorporate environmental or crowd-related factors, limiting their applicability in large-scale environments.

2.2.2 *Crowd Monitoring Systems*

Crowd monitoring approaches in [1, 10] demonstrate that computer vision and deep learning techniques can effectively estimate crowd density and analyze movement patterns with high accuracy in complex scenes. CNN-based models such as those in [11, 12] further improve performance in dense environments; however, they require large training datasets and face challenges in highly occluded scenarios. Survey-based studies in [2, 13] confirm the effectiveness of deep learning methods for crowd analysis, although they do not provide practical implementations. Dataset-oriented work such as [14] contributes standardized benchmarks for training and evaluation but lacks contextual information such as environmental conditions.

More advanced approaches, including transformer-based models in [3], enhance feature extraction and global context understanding in highly congested scenes, but introduce high computational complexity. Similarly, scale-adaptive models in [15] improve performance across varying crowd densities, although they increase training complexity. Real-time crowd monitoring systems such as [16] achieve high accuracy in dynamic environments, but their performance depends on camera quality and infrastructure. Density estimation models such as [17] provide reliable results but face limitations in extremely dense scenarios.

2.2.3 *Heat and Environmental Risk Studies*

Medical and heat-related studies such as [18, 19] highlight the significant impact of high temperatures on human health, confirming that heat exhaustion and heat stroke are critical risks. However, these studies are primarily clinical and reactive rather than real-time or predictive. Environmental research such as [20] further emphasizes heat as a major risk factor but does not involve intelligent or automated systems. Preventive guidelines such as [7] stress the importance of proactive monitoring in mass gatherings, although they remain non-technical. Clinical findings in [6] confirm that early detection of heat stroke is essential to reduce mortality but lack practical implementation mechanisms. Additionally, [21] demonstrates that the combination of heat and crowd density significantly increases health risks in Hajj environments.



2.2.4 Integrated and Smart Systems

Modeling approaches such as [22] provide a useful framework for combining multiple risk factors through weighted models; however, they depend on predefined parameters. Smart system approaches in [23] demonstrate that AI-based early warning systems can improve response time, yet they remain partially integrated. Similarly, [24] presents an IoT-based Hajj monitoring system with effective tracking capabilities but lacks real-time crowd density integration. Context-aware systems in [25] show that integrating multiple data sources enhances adaptability and system intelligence, although it increases system complexity and implementation cost.

2.3 Comparative Analysis

Table 2.1. Crowd Density Estimation Systems

Reference	Year	Methods & Algorithms	Advantage	Limitation	Result	Difference from Darp
[1]	2025	CNN, Computer Vision	High accuracy in dense scenes	Ignores health risks	Accurate detection	Combines density with health and temperature
[11]	2025	CNN	Works well in dense crowds	Requires large datasets	High performance	Uses efficient real-time classification
[17]	2020	ML	Good estimation	Weak in extreme congestion	Moderate accuracy	Improves robustness

This category includes studies that focus on estimating crowd density and counting people in crowded environments using computer vision and deep learning techniques.



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Table 2.2. Real-Time Crowd Monitoring Systems

Reference	Year	Methods & Algorithms	Advantage	Limitation	Result	Difference from Darp
[10]	2025	Deep Learning, Video Analysis	Real-time monitoring	Depends on camera quality	Effective performance	Adds health risk prediction
[16]	2021	AI, CV	Real-time monitoring	Infrastructure dependent	High accuracy	Adds multi-factor analysis

This category includes studies that focus on monitoring crowd movement and congestion in real time to support immediate decision-making and safety management.

Table 2.3. Crowd Deep Learning Architectures

Reference	Year	Methods & Algorithms	Advantage	Limitation	Result	Difference from Darp
[12]	2025	Deep Learning	Good classification	Struggles in extreme congestion	Moderate performance	Uses multi-factor analysis
[3]	2022	Transformer	Strong feature extraction	High complexity	High accuracy	Uses lightweight approach
[15]	2020	Deep Learning	Handles varying densities	Complex training	Improved adaptability	Simplified for real-time

This category includes studies that propose advanced deep learning architectures and models specifically designed for crowd analysis and classification.



Table 2.4. Crowd Surveys and Benchmark Studies

Reference	Year	Methods & Algorithms	Advantage	Limitation	Result	Difference from Darp
[14]	2016	Dataset Processing	Standardized evaluation	Lacks real-world context	Useful training data	Uses real-time contextual data
[2]	2021	Survey	Comprehensive review	No implementation	Analytical insights	Provides practical system
[13]	2021	Survey, DL	High accuracy comparison	No real system	Evaluation only	Applies in real environment

This category includes survey papers and benchmark studies that review, compare, and evaluate existing crowd analysis methods and datasets.



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Table 2.5. Health Monitoring and Symptom-Based Systems

Reference	Year	Methods & Algorithms	Advantage	Limitation	Result	Difference from Darp
[8]	2022	Rule-based AI, NLP	Fast symptom analysis	Depends on user input, no environmental awareness	Quick response	Integrates symptoms with crowd and temperature
[9]	2022	Machine Learning, NLP	Interactive diagnosis	Not real-time, user-dependent	Moderate accuracy	Adds real-time environmental and crowd data
[4]	2023	AI, IoT	Continuous monitoring	Requires wearable devices	Real-time tracking	No wearable dependence
[5]	2022	IoT Sensors	Accurate tracking	Limited scalability	Good monitoring	Uses non-intrusive method

This category includes studies that focus on health monitoring, symptom assessment, and AI-based systems for detecting potential health risks.



Table 2.6. Heat and Environmental Risk Systems

Reference	Year	Methods & Algorithms	Advantage	Limitation	Result	Difference from Darp
[18]	2021	Medical Analysis	Reliable insights	Not real-time	Accurate results	Converts to predictive system
[19]	2024	Statistical Models	Identifies heat risks	No AI integration	Accurate risk detection	Integrates heat with AI system
[20]	1969	Data Analysis	Provides a structured methodology for risk assessment	No intelligent integration	Useful insights	Integrates environment with AI
[7]	2022	Guidelines	Preventive awareness	Not technical	Risk reduction	Converts to automated system
[6]	2020	Medical Models	Early detection	No real-time system	High accuracy	Enables real-time alert
[21]	2024	Data Analysis	Shows combined risks	No solution provided	Risk identification	Provides integrated solution

This category includes studies that focus on environmental factors such as temperature and heat-related conditions that may affect human safety and health.



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Table 2.7. Integrated Intelligent Monitoring Systems

Reference	Year	Methods & Algorithms	Advantage	Limitation	Result	Difference from Darp
[22]	2008	Composite Indicators, Weighted Models	Combines multiple risks	Static weights	Flexible modeling	Practical risk monitoring system
[23]	2024	AI, Rule-based	Fast alerts	Limited integration	Quick response	Uses richer data source
[24]	2023	IoT	Tracks individuals	No full integration	Good tracking	Integrates crowd + health + environment
[25]	2022	AI, Context-aware	Adaptive decisions	High complexity	High performance	Balances efficiency and accuracy

This category includes studies that combine multiple factors such as crowd monitoring, health indicators, and environmental conditions into a unified intelligent monitoring system.



2.4 Identified Gaps and Limitations

Table 2.8. Identified Gaps and Limitations (Part 1)

Reference	Approach Type	Identified Gaps and Limitations
[8], [9]	Digital Health Systems	Rely heavily on user input, lack environmental awareness (temperature/crowd), not suitable for large-scale crowded environments
[1], [10]	Crowd Monitoring (CV/DL)	Focus only on visual data, do not link crowd density to health risks
[18], [19]	Medical / Heat Studies	Clinical and reactive, not designed for real-time or predictive applications
[11], [12]	CNN-based Density Models	Require large datasets, struggle with occlusion in dense scenes
[20]	Environmental Studies	Focus only on temperature, lack integration with intelligent systems
[22]	Composite Models	Depend on predefined/static weights, limited adaptability
[14]	Dataset-based Research	Provide standardized datasets but lack contextual/environmental information
[2], [13]	Survey Studies	Theoretical analysis only, no practical implementation
[3]	Transformer-based Models	High computational complexity, difficult for real-time deployment
[15]	Scale-adaptive Models	Complex training process, require careful tuning



Table 2.9. Identified Gaps and Limitations (Part 2)

Reference	Approach Type	Identified Gaps and Limitations
[4], [5]	AI + IoT Healthcare Systems	Depend on wearable devices, do not consider crowd or environmental factors
[7]	Preventive Guidelines	Non-technical, lack intelligent system implementation
[6]	Clinical Heat Stroke Studies	No real-time implementation mechanisms
[21]	Hajj Risk Studies	Highlight the problem only, no integrated technical solution
[16]	Real-time Crowd Systems	Depend on camera quality and infrastructure
[17]	Density Estimation Models	Face challenges in extremely congested environments
[23]	Smart Early Warning Systems	Lack full integration across domains (health, crowd, environment)
[24]	IoT-based Hajj System	Do not integrate crowd density with health and environmental data
[25]	Context-aware Systems	High system complexity and implementation cost

Key Insight (Research Gap):

Most existing approaches focus on a single aspect (crowd, health, or environment) and lack real-time integration. No system fully combines crowd density, temperature, and health monitoring in a unified real-time framework.



2.5 Project Contribution

To address these limitations, this project proposes an integrated AI-based system that combines:

- CNN-based crowd density estimation for real-time spatial analysis
- Heat risk modeling based on environmental temperature normalization
- Symptom-based user input integration for early health indicators

The system generates a unified risk score using a weighted model and provides real-time, location-based alerts to support decision-making for medical teams.

Unlike existing systems, the proposed solution focuses on:

1. Early detection instead of diagnosis
2. Multi-factor integration instead of isolated analysis
3. Privacy-preserving design without storing personal data
4. Suitability for large-scale environments such as Hajj



Chapter 3

PROBLEM STATEMENT AND OBJECTIVES

3.1 Operational Problem Statement

In the context of the Hajj environment, large-scale mass gatherings face significant health and safety challenges due to high crowd density and extreme temperatures. Existing systems address crowd monitoring using computer vision and deep learning techniques, while others focus on heat-related health risks or symptom-based analysis. However, these systems are often developed independently and lack integration between crowd, environmental, and health-related factors.

As a result, there is a clear gap in the absence of an integrated preventive system that combines crowd density, temperature conditions, and early symptom indicators within a unified framework suitable for highly congested environments such as Hajj.

This leads to problems in early risk detection and fast response, since most current systems focus on diagnosis instead of early warning and may need complex data or devices. This can cause delays in response, weak monitoring of the situation, and increase the risk of heat-related problems during Hajj.

3.2 Functional Goals of the System

The project aims to:

1. Design and develop an intelligent software platform capable of monitoring health risks resulting from crowd congestion and high temperatures during the Hajj season.
2. Employ computer vision techniques to monitor crowd density and identify high-risk zones, targeting a classification accuracy of at least 85% when tested on benchmark datasets, consistent with findings reported in recent literature [1, 10].
3. Provide a hypothetical representation of heat stress risks based on crowd density levels and environmental conditions, using a formula that weights density more heavily (60%) due to its dual impact on both heat stress and mobility restriction, and 40% for normalized temperature.
4. Link crowd density and heat risks to self-reported symptoms through a rule-based engine that dynamically updates zone risk scores, triggering immediate notifications to medical teams when critical levels are detected.
5. Offer a simple user interface that accommodates diverse pilgrim demographics and supports medical teams in decision-making without requiring advanced medical diagnosis, with an end-to-end response time target (from image analysis to alert generation) of under 5 seconds in the prototype system.
6. Enable spatial risk monitoring by dividing Hajj ritual areas into predefined sectors, allowing medical teams to prioritize responses and allocate resources more efficiently.



7. Evaluate the integrated system's performance in a simulated environment using structured scenarios constructed by the project team based on combinations of crowd density levels (low, medium, high) derived from the ShanghaiTech dataset and environmental temperature values obtained from historical data or simulated within realistic Hajj ranges (30 – 45°C). Each scenario is processed using the system's risk model ($0.6 \times \text{Density} + 0.4 \times T_{\text{norm}}$) with rule-based symptom adjustments, and the expected outputs are predefined risk levels (low, preventive warning, high risk, critical), which serve as the ground truth. Scenario validation is ensured by alignment with prior literature and consistency with predefined system thresholds, while performance is assessed using accuracy, precision, recall, and user satisfaction.

The following table is provided to illustrate the evaluation scenarios discussed in this chapter. It presents the combinations of crowd density, temperature, and reported symptoms, along with the expected risk level, to clearly explain how the system validation process is performed.

Table 3.1. Evaluation Scenarios for System Validation

Scenario ID	Crowd Density Level	Temperature (°C)	Symptoms Reported	Ground Truth
S1	Low (1)	30°C	0 Reports	Low
S2	Low (1)	33°C	1-2 Reports	Preventive Warning
S3	Medium (2)	36°C	1-2 Reports	Preventive Warning
S4	Medium (2)	39°C+	3+ Reports	High Risk
S5	High (3)	42°C+	3+ Reports	Critical
S6	High (3)	45°C+	5+ Reports	Critical

Explanation of Table Columns

1) Scenario ID: Represents the identifier of each testing scenario used to evaluate the system under different conditions.

2) Crowd Density Level: Represents the level of crowd congestion detected by the CNN model, classified as Low (1), Medium (2), or High (3).

3) Temperature (°C): Represents the environmental temperature assigned to each scenario, reflecting realistic heat conditions during Hajj.

4) Symptoms Reported: Represents the number of heat-related symptoms reported in each scenario, such as dizziness, fatigue, or discomfort.

5) Ground Truth: Represents the expected final risk classification for each scenario, used as the reference output for system evaluation.



3.3 Scope, Constraints, and Assumptions

This project aims to develop an AI-based prototype to support early health and safety monitoring in crowded and high-temperature environments. The system focuses on analyzing heat- and crowd-related risk indicators using simple user-reported symptoms and general visual cues extracted from images, such as visible fatigue or skin redness. The project does not provide medical diagnosis, does not store personal data, and does not address chronic or complex medical conditions. It is limited to simple and potentially urgent cases related to heat and crowd exposure and will be implemented as a prototype.

Assumptions

The project is based on the following assumptions:

- The Hajj environment can be divided into predefined spatial zones for monitoring.
- Environmental temperature values can be obtained from historical weather data for Makkah or simulated within realistic ranges (30 – 45°C).
- Pilgrims may report simple symptoms through a web-based interface without requiring technical training; however, user participation may be limited in crowded situations. Therefore, symptom reports are considered an additional enhancement rather than a mandatory input.
- The ShanghaiTech Part A dataset provides useful examples of high-density crowd scenes; however, it may not fully represent the unique environmental and behavioral conditions of the Hajj context.



Chapter 4

RESEARCH METHODOLOGY AND PLAN

4.1 System Development Methodology

This project follows an applied development methodology to build a web-based AI prototype for early health risk monitoring in crowded and high-temperature environments such as Hajj. The system integrates computer vision-based crowd density classification, heat-risk representation, and simple self-reported symptoms within predefined spatial zones. The crowd density model is evaluated using accuracy, precision, recall, and F1-score. The dataset is split into 70% training, 15% validation, and 15% testing to ensure proper generalization. Environmental temperature values are obtained from historical weather data for Makkah or simulated within realistic Hajj temperature ranges (30 – 45°C).

4.2 Tools, Platforms, and Technologies

The platform is implemented using Python, with OpenCV for image processing and CNN models developed using TensorFlow or PyTorch to classify density levels (low, medium, high). The output of the CNN-based model is mapped into numerical density levels (Low = 1, Medium = 2, High = 3), which are then used in the risk calculation formula. A lightweight web interface is developed using Flask or Streamlit. Heat-related risk is calculated using a weighted formula:

$$\text{Baseline Risk Score} = 0.6 \times (\text{Density}) + 0.4 \times (T_{\text{norm}}) \quad (4.1)$$

Temperature values are normalized using:

$$T_{\text{norm}} = (T - T_{\text{min}}) / (T_{\text{max}} - T_{\text{min}}) \quad (4.2)$$

This model follows a weighted composite indicator approach as described in the OECD Handbook on Constructing Composite Indicators [22], while temperature contributes 0.4 as an environmental stress factor supported by heat-related health studies [20]. To incorporate health-related indicators, a rule-based symptom engine adjusts the risk score by adding +1 for a single symptom report and +2 for multiple reports (three or more) within the same geographical zone. Self-reported symptoms (e.g., dizziness, fatigue, itching) are collected through the system interface and used only to adjust the risk score without storing personal information. The final score is categorized into predefined alert levels (low, medium, high, or critical), which are displayed through an administrative monitoring interface.



4.3 Data Sources

The crowd density model is trained using the publicly available ShanghaiTech Part A dataset [14], which contains high-density crowd scenes representative of large-scale gatherings such as Hajj. The dataset includes a variety of crowd images with different levels of congestion, perspectives, and occlusions, making it suitable for training and evaluating the crowd density classification model. The dataset is also used to support the classification of crowd density into three levels (Low, Medium, High), which are required for the risk assessment process.

If additional data is required, ShanghaiTech Part B or the UCF-QNRF dataset can be used as supplementary datasets to improve model generalization across different crowd densities.

In addition to crowd data, environmental temperature values are incorporated into the system. Due to the lack of access to real-time temperature data in the Hajj environment and the constraints of the project, simulated temperature values are used. These values are generated within a realistic range of 30 – 45°C, reflecting typical weather conditions during the Hajj season in Makkah.

4.4 Ethical Considerations

The proposed system adheres to internationally recognized ethical standards in artificial intelligence, data governance, and public health. Given the sensitive context of mass gatherings such as Hajj, the design and implementation of this project prioritize privacy, fairness, transparency, and responsible use of technology.

4.4.1 Privacy and Data Protection

The system does not collect, store, or process personally identifiable information. All analyses rely on publicly available datasets, such as the ShanghaiTech dataset, ensuring the protection of individual privacy and compliance with ethical data practices.

4.4.2 Responsible and Non-Diagnostic Use

The proposed solution serves as an early risk detection and decision-support tool. It does not provide medical diagnoses or replace professional healthcare services. Instead, it generates preventive alerts to support medical teams and enhance pilgrim safety.

4.4.3 Transparency and Explainability

The methodologies, algorithms, and data sources used in the system are clearly documented to ensure transparency and accountability. This promotes trust in AI-driven decision-making and aligns with global ethical AI principles.

4.4.4 Fairness and Bias Mitigation

The system utilizes standardized benchmark datasets and validated evaluation metrics to minimize bias and ensure equitable performance across diverse populations. Continuous evaluation and testing are conducted to maintain reliability and fairness.



4.4.5 Data Security and Confidentiality

Any environmental or symptom-related data are handled securely and used solely for risk assessment. No personal or medical data are stored or shared, ensuring confidentiality and compliance with ethical standards.

4.4.6 Compliance with International and National Ethical Frameworks

The project aligns with globally recognized AI ethics principles, including those established by UNESCO, the OECD, and the European Commission, as well as national guidelines issued by the Saudi Data and Artificial Intelligence Authority (SDAIA).

The proposed system complies with international ethical standards in artificial intelligence and data governance, aligning with guidelines from UNESCO, OECD, the European Commission, SDAIA, and the World Health Organization [7, 26–29].

4.5 Risk Assessment

To ensure the success of the project, several potential risks have been identified with appropriate mitigation strategies. Technically, the main risk is that the 85% target accuracy for the crowd model may not be achieved. To address this, pretrained models will be used along with data augmentation techniques and hyperparameter tuning. Another risk is that the dataset may not adequately represent actual congestion conditions in the Hajj environment, in which case additional datasets such as UCF-QNRF will be used. Regarding environmental data, the lack of accurate temperature measurements for specific pilgrimage areas is a potential challenge. This will be addressed by simulating temperature values within realistic ranges (30 – 45°C) based on historical weather data for Mecca. Integration challenges between different system units may also arise; therefore, a modular design approach will be adopted with early integration testing. On the other hand, difficulty in achieving the target response time of under 5 seconds may be faced due to model complexity. This will be resolved by optimizing model efficiency and reducing input image size while maintaining accuracy. Finally, limited access to high-performance computational resources will be mitigated by using cloud-based platforms such as Google Colab and Kaggle.



4.6 Milestone-Based Project Schedule

Table 4.1. Project Timeline

Week	Phase	Activities/Deliverable
1	Project Initiation	Define project scope, objectives, and team roles
2	Background Study	Study Hajj health risks, crowd density estimation, and heat stress concepts
3	Literature Search	Collect research papers related to CNN crowd analysis, heat risk, and symptom-based systems
4	Literature Review	Review and summarize literature and identify research gap (Project Proposal)
5-6	Project Planning	Prepare timeline, milestones, task allocation, and risk assessment (Project Management Plan)
7	Report Writing	Write GP-I Report Chapters 1-2 (Introduction, Background)
8	Requirements	Document functional & non-functional requirements, risk formula (System Requirements)
9	Report Writing	Write GP-I Report Chapters 3-4 (Problem Statement, Methodology)
10	Report Writing	Write GP-I Report Chapter 5 (Requirements, Use-Case)
11	Complete Chapter 5	System architecture, algorithms, design decisions, testing strategy
12	Report Review	Review and refine all chapters
13	Report Finalization	Final formatting and proofreading
14	Submission	Prepare final GP-I Progress Report
15	Presentation Preparation	Prepare slides and rehearse oral presentation
16	Final Presentation	Present proposal, planning, and progress (GP-I Oral Presentation)



Milestone Description

This project is divided into a set of clearly defined milestones to ensure systematic development and timely completion of the proposed system.

Milestone 1: Project Planning and Requirement Definition This milestone focuses on defining the project scope, understanding the problem domain, and identifying functional and non-functional requirements. The project objectives are finalized to ensure alignment with the intended outcomes. (Weeks 1-2)

Milestone 2: Literature Review and Gap Identification Relevant studies related to crowd density estimation, heat-related health risks, and symptom-based health systems are reviewed. Existing approaches are analyzed to identify limitations and research gaps addressed by this project. (Weeks 3-4)

Milestone 3: Project Proposal Submission The project proposal is completed, including background, scope, description, objectives, timeline, expected outcomes, methodology, and references. (Week 4)

Milestone 4: Project Management Plan Submission The project management plan is prepared, including timeline, milestones, task allocation, and risk assessment. (Weeks 5-6)

Milestone 5: Report Writing (Chapters 1-2) The GP-I report chapters are written, including Introduction and Motivation (Chapter 1) and Background and Related Work (Chapter 2). (Week 7)

Milestone 6: System Requirements and Design Submission Functional and non-functional requirements are documented. (Week 8)

Milestone 7: Report Writing (Chapters 3-4) The GP-I report chapters are written, including Problem Statement and Objectives (Chapter 3) and Methodology and Project Plan (Chapter 4). (Week 9)

Milestone 8: Report Writing (Chapter 5 - Part 1) The GP-I report Chapter 5 is written, including Functional Requirements, Non-Functional Requirements, and Use-Case Analysis. (Week 10)

Milestone 9: Chapter 5 Completion The remaining sections of Chapter 5 are completed, including System Architecture, High-Level Algorithms and Models, Design Decisions and Justification, and Planned Testing and Evaluation Strategy. (Week 11)

Milestone 10: Report Review and Finalization All chapters are reviewed, refined, and formatted. Final proofreading is completed. (Weeks 12-13)



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Milestone 11: GP-I Progress Report Submission The final GP-I Progress Report is prepared and submitted. (Week 14)

Milestone 12: GP-I Oral Presentation Presentation slides are prepared, and the oral presentation is delivered to the supervisor and examiner. (Weeks 15-16)



Chapter 5

SYSTEM ANALYSIS AND SYSTEM DESIGN

5.1 Functional Requirements

5.1.1 *Crowd Density Classification*

The system shall analyze crowd images and classify density levels as Low, Medium, or High using a CNN-based model to identify overcrowded areas and support early preventive actions to reduce risks.

5.1.2 *Zone-Based Monitoring*

Target Users: Authorities

The system shall organize the monitored area into predefined zones (such as Zone A and Zone B), enabling authorities to track risk levels for each location and prioritize high-risk zones for timely intervention.

5.1.3 *Temperature Retrieval and Normalization*

The system shall retrieve environmental temperature values from historical weather data or simulated inputs and normalize them before including them in the risk calculation process.

5.1.4 *Baseline Risk Score Calculation*

The system shall calculate a baseline risk score by combining crowd density levels with normalized temperature values using a predefined weighted formula.

5.1.5 *Symptom Reporting Interface*

Target Users: Pilgrims

The system shall allow pilgrims to report simple symptoms, such as dizziness, fatigue, or itching, through a user-friendly web interface, supporting early detection of potential health risks and increasing awareness.

5.1.6 *Rule-Based Risk Adjustment*

The system shall dynamically adjust the risk score of each zone based on the number of symptoms reported within the same area to provide accurate and real-time risk assessment.



5.1.7 Alert Generation and Dashboard Visualization

Target Users: Medical Teams, Authorities, and Pilgrims

The system shall generate alert levels (Low, Preventive Warning, High Risk, or Critical) and display them on an administrative dashboard for medical teams and authorities. Additionally, the system shall provide preventive notifications to pilgrims in high-risk zones.



5.2 Non-Functional Requirements

5.2.1 Performance

The system shall process an image and generate an alert within 5 seconds from the time of image analysis to alert notification under normal operating conditions, measured by logging the start and end time using a Python script (`time.time()`).

5.2.2 Reliability

The system shall achieve 99% uptime with no more than one unexpected failure per 1,000 user requests during continuous 8-hour testing periods, measured by monitoring system logs and calculating $(\text{uptime} / \text{total time}) \times 100$ using a Python script or Google Colab.

5.2.3 Scalability

Due to limited student resources (no access to 500 real users), the system shall maintain a response time under 5-7 seconds when handling up to 500 concurrent virtual users using Apache JMeter (100, 200, 300, 400, then 500 users with a gradual ramp-up).

5.2.4 Ease of Use

The system shall allow pilgrims to report a symptom within no more than two clicks without requiring technical training, measured by user testing with 5 participants (fellow students) and counting the number of clicks required to complete the task.

5.2.5 Security and Privacy

The system shall store no personally identifiable information (PII) and aggregate all symptom reports at the zone level only during all operational phases, verified by inspecting the database tables to confirm that no names, IDs, or contact information are stored. In addition, the administrative dashboard shall be accessible only to authorised medical teams and authorities through secure access credentials, preventing unauthorised access by pilgrims or other users.

5.2.6 Maintainability

The system shall be designed with at least four independent modules (crowd density, heat simulation, symptom engine, web interface) without affecting the others when one module is updated, verified by demonstrating that each module can be modified or replaced independently during testing.

5.2.7 Availability

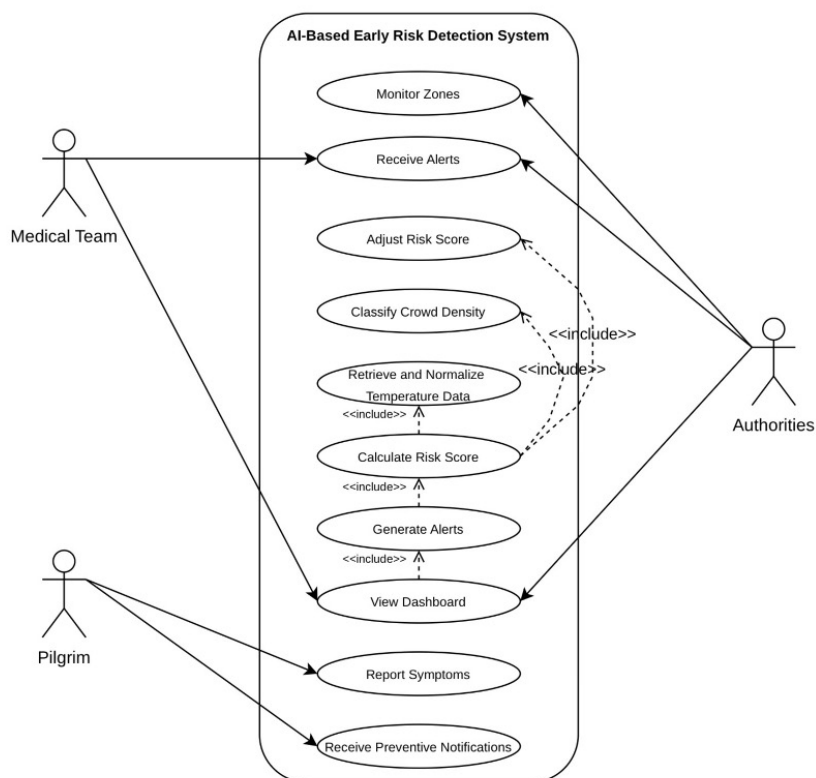
The system shall maintain at least 99% availability during 24-hour simulation periods, measured by tracking total system uptime and downtime using a Python script or Google Colab, then calculating $(\text{uptime} / \text{total time}) \times 100$.



5.3 Use-Case Analysis

The use case diagram illustrates the interaction between the user's pilgrims, medical teams, and authorities and the proposed AI-based early risk detection system. Pilgrims can report symptoms and receive preventive notifications, while medical teams and authorities monitor system outputs through the dashboard and receive alerts for high-risk zones. The system processes crowd density, temperature data, and user-reported symptoms to calculate risk levels and generate alerts, supporting timely decision-making in crowded environments such as Hajj.

Figure 5.1. Use Case Diagram





5.4 User Scenarios

5.4.1 Scenario 1: Pilgrim Reporting Symptoms

Actor: Pilgrim

Goal: Report heat-related symptoms anonymously.

Preconditions:

- The pilgrim has access to the web-based system.
- The system is operational.
- The pilgrim is located within a defined zone (e.g., Zone A).

Main Flow:

1. The pilgrim opens the symptom reporting page.
2. The pilgrim selects the current zone.
3. The pilgrim selects symptoms from a predefined list (e.g., dizziness, fatigue).
4. The pilgrim submits the report.
5. The system records the data anonymously.
6. The system associates the report with the selected zone.
7. The system updates the risk score based on accumulated reports.
8. If the risk level increases, a preventive notification is sent to the pilgrim.

Alternative Flow:

- If the pilgrim does not select a zone → the system prompts the user to select a zone before submission.
- If the user submits an empty form → the system displays a validation message.

Postcondition: The symptom report is stored anonymously and contributes to the zone-level risk assessment.

5.4.2 Scenario 2: High-Risk Zone Monitoring by Authorities

Actor: Authorities

Goal: Monitor zones and detect high-risk conditions in real-time.

Preconditions:

- The authority has access to the system dashboard.
- Cameras and temperature sensors are active.
- The system is processing incoming data.

**Main Flow:**

1. The authority opens the administrative dashboard.
2. The system displays real-time data for all zones.
3. The system analyzes incoming images using the CNN model.
4. The system classifies crowd density (Low, Medium, High).
5. The system collects and processes temperature data.
6. The system calculates the baseline risk score for each zone.
7. If a zone reaches a high or critical level, the system generates an alert.

Alternative Flow:

- If image data is unavailable → the system continues using temperature and symptom data only.
- If sensor data is delayed → the system uses the most recent available data.

Postcondition: High-risk zones are identified, and alerts are generated for timely intervention.

5.4.3 Scenario 3: Medical Team Responding to Alert

Actor: Medical Team

Goal: Respond quickly to critical situations and reduce health risks.

Preconditions:

- The medical team has access to the system dashboard.
- The system is actively monitoring zones.
- An alert is triggered due to high risk.

Main Flow:

1. The medical team monitors the dashboard during peak hours.
2. The system detects high crowd density, high temperature, and multiple symptom reports.
3. The system calculates the risk score and exceeds the critical threshold.
4. The system sends an alert notification.
5. The medical team receives the alert.
6. The team identifies the affected zone (e.g., Zone B).
7. The team takes appropriate action (e.g., dispatching assistance, guiding pilgrims).

Alternative Flow:

- If the alert is not acknowledged → the system re-sends the notification.
- If multiple zones are critical → alerts are prioritized based on risk level.



Postcondition: The medical team is informed and takes action to handle potential heat-related cases.

5.5 Entity-Relationship Diagram (ER Diagram)

The database consists of three main entities: zones, symptoms, and alerts. Pilgrims interact with the system as external actors (as shown in the Use-Case Diagram) but their personal data is not stored in the database to preserve privacy. Therefore, no Pilgrim table in the ER diagram or database schema.

Figure 5.2. ER Diagram

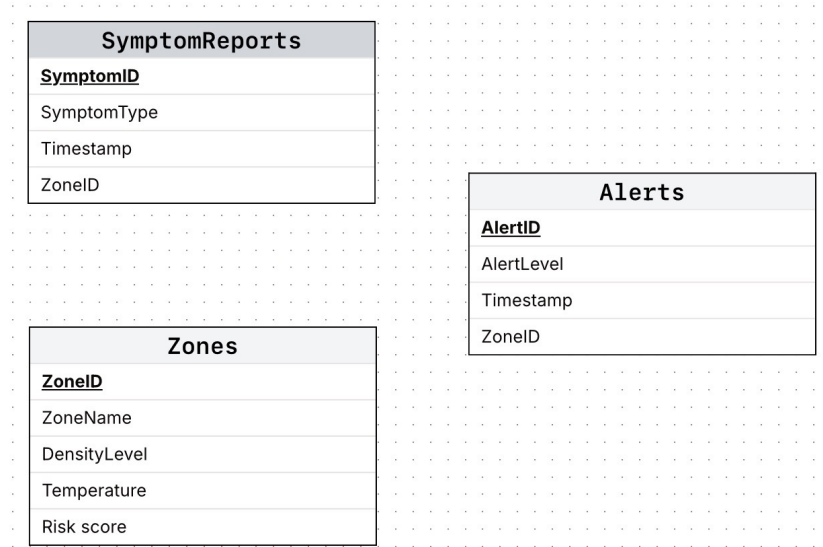


Figure 5.3. Relation Schema

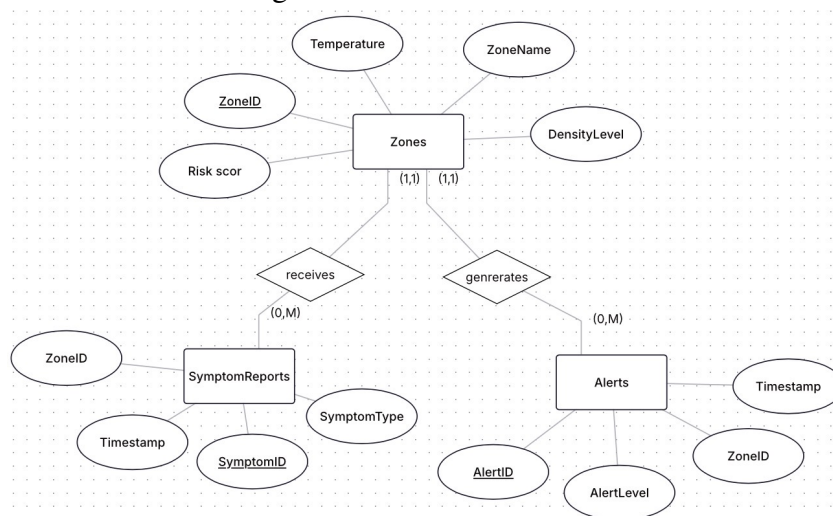




Table 5.1. Relationships and Cardinalities

Relationship	Cardinality	Description
Zones → Symptoms	(0,M)	A zone can have zero or multiple symptom reports
Symptoms → Zones	(1,1)	A symptom report belongs to exactly one zone
Zones → Alerts	(0,M)	A zone can trigger zero or multiple alerts
Alerts → Zones	(1,1)	An alert belongs to exactly one zone

5.6 Database Schema

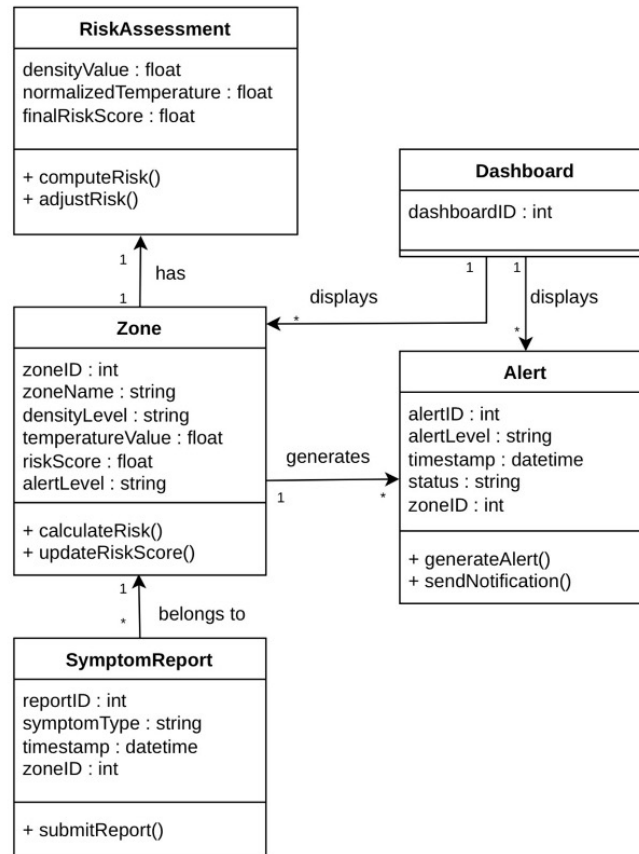
Table 5.2. Database Schema

Table Name	Column Name	Data Type	Description
Zones	ZoneID	INT	Unique zone identifier
	ZoneName	VARCHAR(10)	Zone name A, B, C
	DensityLevel	INT	1=Low, 2=Medium, 3=High
	Temperature	FLOAT	Temperature
	RiskScore	FLOAT	Calculated risk score
SymptomReports	SymptomID	INT	Unique symptom report identifier
	SymptomType	VARCHAR(20)	dizziness, fatigue, itching
	ZonesID	INT	Foreign key
	Timestamp	DATETIME	Time of symptom report
Alerts	AlertID	INT	Unique alert identifier
	AlertLevel	VARCHAR(20)	Low, Warning, High, Critical
	Timestamp	DATETIME	Time alert was generated
	Status	VARCHAR(20)	Sent, Acknowledged
	ZoneID	INT	Foreign key references zones table



5.7 Class Diagram

Figure 5.4. Class Diagram



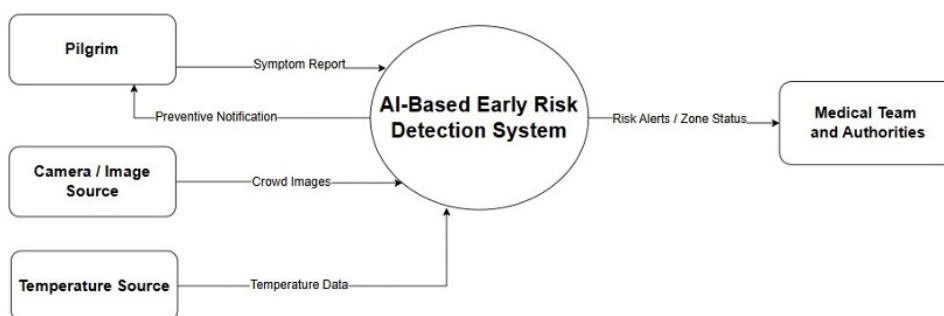
This class diagram illustrates the main classes of the proposed Darp system and their relationships. The Zone class stores zone-related information such as density level, temperature value, risk score, and alert level. Each Zone is associated with a RiskAssessment object used to compute and adjust risk scores. SymptomReport objects belong to specific zones and contribute to the risk evaluation process. Based on the calculated risk level, a Zone can generate an Alert, which contains alert details and notification functions. The Dashboard class displays zone information and generated alerts to support monitoring and decision-making.



5.8 Data Flow Diagram

5.8.1 Level 0 DFD (Context Diagram)

Figure 5.5. Data Flow Diagram Level 0 - Context Diagram



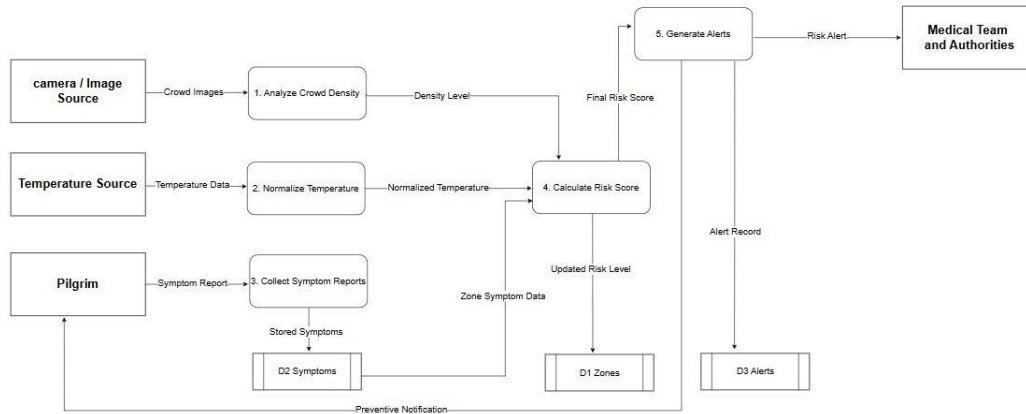
This diagram represents the Level 0 Data Flow Diagram, also known as the Context Diagram, for the AI-Based Early Risk Detection System. It shows the overall interaction between the system and the external entities without presenting the internal processes.

The Pilgrim sends a Symptom Report to the system and receives a Preventive Notification in return. The Camera or Image Source provides Crowd Images, while the Temperature Source sends Temperature Data. All these inputs are processed by the system internally. After analyzing the received data, the system sends Risk Alerts and Zone Status information to the Medical Team and Authorities. Therefore, this diagram provides a high-level view of how data enters and leaves the system and how the system communicates with external actors.



5.8.2 Level 1 DFD

Figure 5.6. Data Flow Diagram Level 1



This diagram represents the Level 1 Data Flow Diagram (DFD) for the AI-Based Early Risk Detection System. It illustrates how the system operates internally by dividing it into a set of interconnected processes that work together to transform input data into meaningful outputs.

The system begins by receiving inputs from external entities. The Camera or Image Source provides crowd images to the **Analyze Crowd Density** process, which determines the crowd density level. At the same time, the Temperature Source sends temperature data to the **Normalize Temperature** process, where it is converted into a normalized value suitable for further processing. In addition, the Pilgrim sends a Symptom Report to the **Collect Symptom Reports** process, where this information is stored in the Symptoms Database (D2 Symptoms).

The outputs from these processes, including Density Level, Normalized Temperature, and Zone Symptom Data, are then forwarded to the core process, **Calculate Risk Score**. This process integrates all inputs to compute the final risk level for each zone.

Once the risk level is calculated, it is stored in the Zones Database (D1 Zones). The Final Risk Score is then passed to the **Generate Alerts** process, which is responsible for producing alerts based on the calculated risk level.

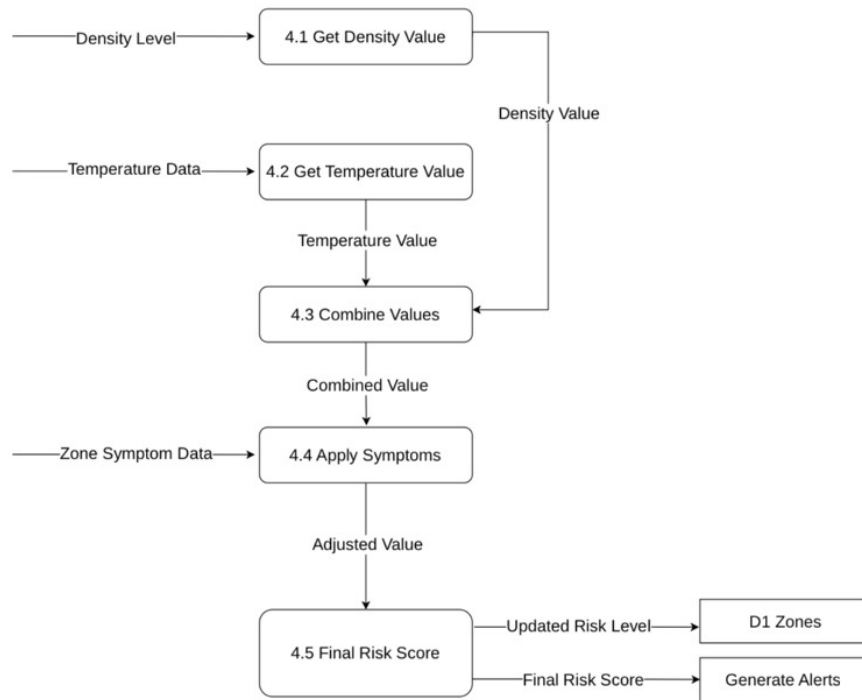
The generated alerts are stored in the Alerts Database (D3 Alerts) as Alert Records. At the same time, Risk Alerts are sent to the Medical Team and Authorities to support decision-making and enable timely intervention. Additionally, Preventive Notifications are sent to the Pilgrim to help reduce exposure to potential risks.

Overall, this diagram demonstrates how the system integrates multiple data sources, including visual data, environmental data, and user-reported symptoms, to generate effective alerts and support early risk detection in crowded environments.



5.8.3 Level 2 DFD - Calculate Risk Score Process

Figure 5.7. Data Flow Diagram Level 2 - Calculate Risk Score



This diagram represents the Level 2 Data Flow Diagram (DFD) of the Calculate Risk Score process in the AI-Based Early Risk Detection System. It illustrates how the risk calculation process is decomposed into smaller steps, where data is processed gradually until the final result is produced.

The process begins by receiving inputs from Level 1. The Density Level is sent to **4.1 Get Density Value**, where it is converted into a usable density value. At the same time, Temperature Data is sent to **4.2 Get Temperature Value**, where it is transformed into a temperature value suitable for processing.

The outputs of these two processes, Density Value and Temperature Value, are then passed to **4.3 Combine Values**, where they are integrated to produce a combined value that reflects the impact of both crowd density and temperature on the risk level.

Next, the combined value is sent to **4.4 Apply Symptoms**, which also receives Zone Symptom Data (from Level 1). In this step, the system adjusts the risk level based on reported symptoms, making the evaluation more accurate and context aware.

Finally, the adjusted value is passed to **4.5 Final Risk Score**, which represents the last stage of the process, where the system generates the final risk score for each zone.



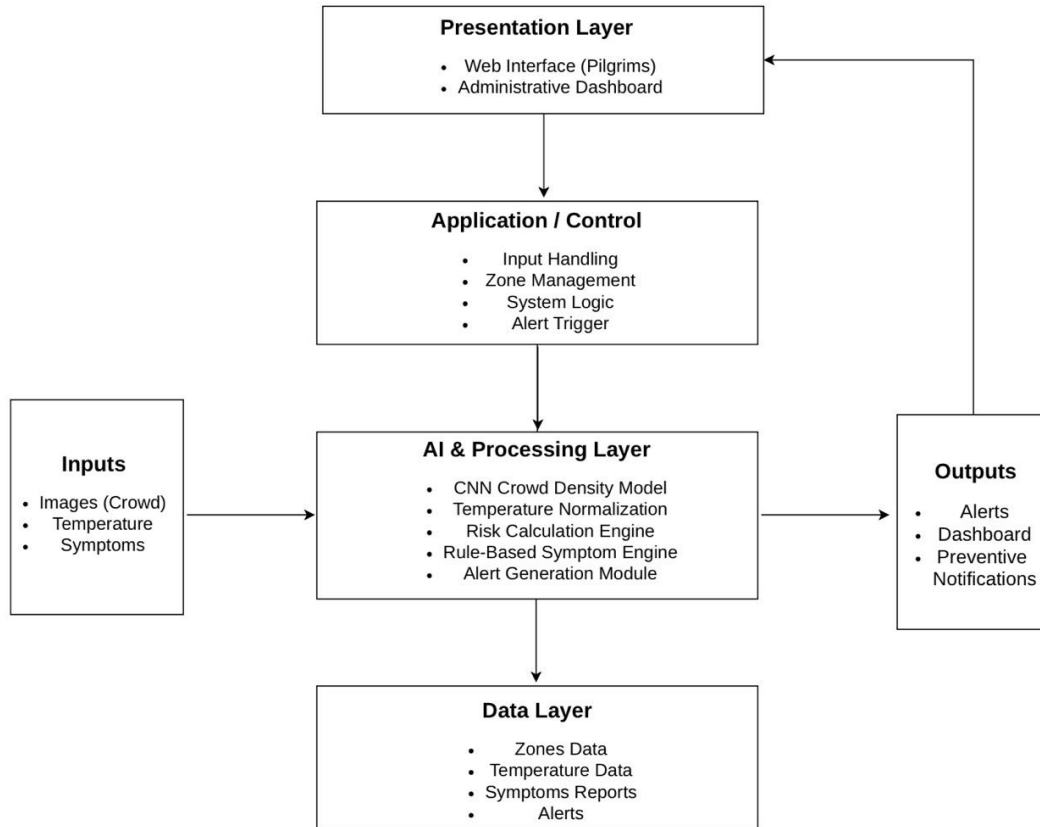
The result is then used in two ways: it is stored in D1 Zones as an Updated Risk Level, and it is also sent to the Generate Alerts process as the Final Risk Score, enabling the system to produce alerts and support timely decision-making.

Overall, this diagram demonstrates how the system integrates crowd density, temperature, and symptom data to generate an accurate and dynamic risk assessment.



5.9 System Architecture

Figure 5.8. System Architecture



This figure illustrates the overall System Architecture of the proposed system, which is designed using a Layered Architecture approach. This architectural style was selected because it divides the system into multiple organized layers, where each layer is responsible for a specific function within the system. Such separation improves system clarity, simplifies development and maintenance, allows smooth data flow between system components, and enhances scalability, making future improvements easier to implement.

Presentation Layer

At the top is the Presentation Layer, which allows users to interact with the system through a web interface for pilgrims and an administrative dashboard for medical teams and authorities, where they can submit information, monitor zone status, and view outputs such as alerts, risk levels, and notifications.



Application / Control Layer

Below it is the Application/Control Layer, which manages system workflow by handling incoming requests, organizing monitoring zones, and applying system rules before directing data for processing or presentation.

AI & Processing Layer

The AI & Processing Layer receives the main inputs (crowd images, temperature data, and symptom reports) and analyzes them through the CNN Crowd Density Model, Temperature Normalization, Risk Calculation Engine, and Rule-Based Symptom Engine to produce the final risk level and generate alerts.

Data Layer

At the bottom is the Data Layer, which stores processed system information such as zone data, temperature records, symptom reports, and generated alerts for future access and monitoring.

On the left side are the Inputs, while on the right side are the Outputs, representing the final alerts and dashboard results and preventive notifications. Overall, this layered design ensures efficient and structured data flow throughout the system.



5.10 High-Level Algorithms and Models

The system is based on two main models of risk analysis and classification:

5.10.1 First: Crowd Density Classification Model

A Convolutional Neural Network (CNN) is used to analyze images taken from different Hajj areas. It performs the following:

- Receives an image of a specific area (e.g., region A or B)
- Analyzes the visual features of the image to estimate the level of crowding
- Classifies the region into one of three categories: Low Density, Medium Density, or High Density

The model will be trained using the ShanghaiTech Part A dataset, a collection containing high-density crowd scenes similar to the Hajj environment, targeting a classification accuracy of at least 85%.

Based on [1, 10] where similar CNN models achieved 87 – 90% accuracy. Setting it at 85% is realistic for a student prototype and ensures meaningful risk detection without overfitting.

5.10.2 Second: Risk Calculation and Adjustment Model

After obtaining the density level, the system calculates the risk score using a weighted formula:

$$\text{Baseline Risk Score} = (\text{Density} \times 0.6) + (\text{Normalized Temperature} \times 0.4) \quad (5.1)$$

Where:

- Density: numerical value (1 = low, 2 = medium, 3 = high)
- Normalized temperature: a value between 0 and 1 calculated by:

$$T_{\text{norm}} = (T - T_{\text{min}}) / (T_{\text{max}} - T_{\text{min}}) \quad (5.2)$$

Where $T_{\text{min}} = 30^{\circ}\text{C}$ and $T_{\text{max}} = 45^{\circ}\text{C}$ (typical Hajj temperature range) [18, 19].

Density has been given a higher weight (60%) because it has a dual effect: it increases heat stress and makes movement and ventilation more difficult [22], and the 40% temperature is supported by heat-related health studies [7, 20].



5.10.3 Symptom-Based Risk Adjustment (Rule-Based Engine)

After calculating the baseline risk score, the rule-based engine adjusts the score based on reports received from pilgrims in the same area:

- +1 for a single symptom report (dizziness, fatigue, itching)
- +2 for three or more reports in the same region

Based on mass-gathering health studies [8,9,18,21], the rule-based approach keeps the system lightweight (not medical diagnosis).

5.10.4 Final Risk Score and Alert Levels

Table 5.3. Final Risk Score and Alert Levels

Risk Score	Alert Level	Action
1.0 – 1.5	Low	No action
1.6 – 2.0	Preventive Warning	Send notification to pilgrims
2.1 – 2.6	High Risk	Send strong warning
> 2.6	Critical	Immediate notification to medical teams

The score range (1.0-3.0) was divided into four equal intervals, each of width 0.5 using the equal-interval classification method. This method is a standard statistical technique widely used in public health warning systems [30]. The four levels follow standard public health warning systems (Low → Warning → High → Critical) [6, 7, 23].



5.10.5 Pseudocode for the Complete Risk Calculation Process

INPUT: crowd_image, temperature, symptom_reports (list)

OUTPUT: risk_level, alert

```
1. density_category <- CNN_classify(crowd_image) // Low, Medium, or High
2. density_value <- map_to_numerical(density_category) // 1, 2, or 3
3. T_norm <- (temperature - 30) / (45 - 30) // Normalize to [0,1]
4. baseline_risk <- (density_value * 0.6) + (T_norm * 0.4)
5. symptom_count <- count_reports_in_zone(symptom_reports)
6. IF symptom_count == 1 THEN
    adjustment <- +1
  ELSE IF symptom_count >= 3 THEN
    adjustment <- +2
  ELSE
    adjustment <- 0
7. final_risk <- baseline_risk + adjustment
8. IF final_risk <= 1.5 THEN
    risk_level <- "Low"
    alert <- "No action"
  ELSE IF final_risk <= 2.0 THEN
    risk_level <- "Preventive Warning"
    alert <- "Send notification to pilgrims"
  ELSE IF final_risk <= 2.6 THEN
    risk_level <- "High Risk"
    alert <- "Send strong warning"
  ELSE
    risk_level <- "Critical"
    alert <- "Immediate notification to medical teams"
9. RETURN risk_level, alert
```



5.11 Design Decisions and Justification

This section presents the key design decisions made in developing the proposed system, along with clear justifications based on the project requirements, identified research gaps, and practical constraints.

5.11.1 Use of CNN for Crowd Density Classification

A Convolutional Neural Network (CNN) is used to classify crowd density levels (Low, Medium, High).

Justification:

CNNs are widely recognized for their strong performance in image analysis and crowd density estimation, especially in complex and dense environments. Compared to traditional image processing techniques, CNNs can automatically extract spatial features and handle occlusion more effectively. This decision directly addresses the need for accurate real-time crowd analysis, which is essential for risk assessment in crowded environments like Hajj. Additionally, CNN-based models were shown in prior studies to achieve high accuracy in dense scenes, making them suitable for this application.

5.11.2 Multi-Factor Risk Integration

The system integrates three main factors:

- Crowd density
- Environmental temperature
- User-reported symptoms

Justification:

Most existing systems focus on a single aspect (either crowd, health, or environment), leading to incomplete risk assessment. This limitation was clearly identified as a research gap in the literature review. By combining these factors into a unified model, the system provides a more comprehensive and realistic understanding of risk, improving early detection and decision-making.

5.11.3 Weighted Risk Formula

The baseline risk score is calculated using a weighted formula that assigns 60% to crowd density and 40% to temperature.

Justification:

The weighting approach follows composite indicator methodologies, where multiple factors are assigned different importance levels based on their impact [22]. Temperature is supported in the literature as a key environmental risk factor in heat-related conditions, especially in mass gatherings such as Hajj [7]. Crowd density was assigned a higher weight because it has a dual effect: it increases heat stress and restricts movement and ventilation, making it a more critical factor in real-world risk scenarios. Therefore, the selected weights (0.6 and 0.4) were defined as a design decision to better reflect the combined impact of environmental and physical factors on risk.



5.11.4 Use of Rule-Based Engine for Symptoms

A rule-based system is used to adjust risk scores based on symptom reports.

Justification:

Instead of using complex medical AI models, a rule-based approach was chosen because:

- The system is designed for early warning, not medical diagnosis
- It reduces computational complexity
- It improves interpretability and transparency

This aligns with the project goal of providing a lightweight and practical prototype while still incorporating health-related indicators effectively.

5.11.5 Zone-Based System Design

The monitored environment is divided into predefined zones (e.g., Zone A, Zone B).

Justification:

Crowded environments such as Hajj are spatially complex and require more organized monitoring. A zone-based approach enables localized risk detection by linking data to specific areas, which helps medical teams respond more quickly and allocate resources more effectively. This design improves system scalability and makes the output more actionable compared to global analysis.

5.11.6 Use of Simulated Temperature Data

The system uses simulated temperature values generated randomly within a realistic range (30 – 45°C) instead of real-time data sources.

Justification:

This approach is supported by research, as simulation is commonly used to test algorithms and evaluate systems without requiring real-world deployment [31]. In addition, simulation often relies on random values to model real-world uncertainty and generate diverse scenarios, as highlighted in domain randomization approaches [32]. Therefore, using random temperature values is a valid and effective method for testing and evaluating the system.

5.11.7 Web-Based Lightweight Architecture

The system is implemented as a lightweight web-based application using frameworks such as Flask or Streamlit.

Justification:

A web-based solution ensures easy accessibility for different users, including pilgrims, authorities, and medical teams, without requiring installation. It also allows faster deployment and testing of the system and adopting a lightweight architecture supports the system requirement of achieving a fast response time of under five seconds, which is essential for real-time alert generation in critical situations.



5.11.8 Privacy-Preserving Design

The system does not store any personal or identifiable user data.

Justification:

Privacy is a critical concern in large-scale public systems, especially in sensitive environments such as Hajj. By avoiding the storage of personal data, the system minimizes ethical risks, ensures compliance with data protection standards, and increases user trust. This design choice aligns with internationally recognized AI ethics and data governance principles, which emphasize privacy, transparency, and responsible data usage [7, 26–29].

5.11.9 Use of Public Dataset (ShanghaiTech Part A)

The ShanghaiTech Part A dataset is selected as the primary dataset for training the crowd density model. Although the dataset is not collected from the Hajj environment, it provides high-density crowd scenes that are similar in terms of congestion levels. Due to the lack of publicly available Hajj-specific datasets, ShanghaiTech is used as a reliable benchmark dataset.

Justification:

Using a publicly available dataset saves time and effort compared to collecting custom data. In addition, the dataset is widely used in research and provides diverse crowd scenarios with varying levels of density, which improves the model's ability to generalize. The presence of high-density crowd scenes similar to real-world environments such as Hajj makes it suitable for training and evaluation.

5.11.10 Modular System Design

The system is designed using a modular structure, separating components such as vision processing, risk calculation, and user interface.

Justification:

This design makes the system easier to manage because each part works independently and can be updated without affecting the others. It also helps in finding and fixing errors more quickly. In addition, it allows the system to be expanded in the future by adding new features easily. This is useful in AI systems, where models may need updates or improvements over time.



5.12 Planned Testing and Evaluation Strategy

Table 5.4. Planned Testing and Evaluation Strategy (Part 1)

Test Category	Description	Metric / Criteria	Measurement Method / Tool
Functional Testing	Verify correct operation of all system components (density classification, zones, risk calculation, alerts)	Correct classification (Low/Medium/High), correct alert level	Manual verification + comparison with expected outputs
Model Evaluation	Evaluate CNN crowd density model performance	Accuracy $\geq 85\%$, Precision, Recall, Confusion Matrix	Python (TensorFlow/PyTorch), Scikit-learn
Risk Evaluation (Scenarios)	Evaluate risk level classification using predefined scenarios	Match between predicted and expected risk levels (Ground Truth)	Reference scenarios based on density + temperature combinations
Performance	Ensure fast system response	Alert generated ≤ 5 seconds	Python timing (time.time())
Scalability	Test system under increasing number of users	Response time remains acceptable ($< 5-7$ sec) with up to 50-500 users	Apache JMeter
Reliability	Ensure stable system operation	$\geq 95-99\%$ uptime, minimal failures	Log monitoring + uptime calculation



Table 5.5. Planned Testing and Evaluation Strategy (Part 2)

Test Category	Description	Metric / Criteria	Measurement Method / Tool
Availability	Measure system availability over time	$\geq 99\%$ availability	Uptime tracking (Python / Colab)
Usability	Evaluate ease of use for pilgrims and staff	Task completed within ≤ 2 clicks, user satisfaction	User testing (5 participants)
Security & Privacy	Ensure no sensitive data storage	No PII stored	Database inspection
Maintainability	Ensure modular system design	Independent modules (4 modules)	Module testing and isolation
End-to-End Testing	Validate full system workflow	Correct risk calculation + alert + dashboard output	Simulated real-world scenarios



5.13 Challenges

5.13.1 *Lack of Real-Time Temperature Data in Hajj Environment*

One of the main challenges in this project was the lack of access to real-time temperature data in the Hajj environment from reliable sources or sensors. This limitation made it difficult to incorporate accurate environmental conditions into the system. As a result, simulated temperature values were used within a realistic range ($30 - 45^{\circ}\text{C}$) to represent the high-temperature conditions during Hajj. While this approach allows effective testing and system evaluation, it does not fully reflect real-time variations, which may affect the accuracy of risk assessment in real-world Hajj scenarios.

5.13.2 *Limited Availability of Hajj-Specific Crowd Data*

Another challenge faced in this project was the limited availability of crowd datasets collected directly from the Hajj environment. Most publicly available datasets are designed for general crowd analysis and may not fully represent the unique movement patterns, environmental conditions, and crowd behavior during Hajj. Therefore, the ShanghaiTech Part A dataset was used because it contains highly congested crowd scenes similar to large scale gatherings. However, the differences between the dataset environment and real Hajj conditions may affect the system's ability to fully generalize in real-world deployment scenarios.

5.13.3 *Limited Reference Systems*

Another challenge in this project was the lack of integrated reference systems that combine crowd monitoring, environmental conditions, and health-related analysis within a single framework. Most existing systems focus on only one aspect, making it difficult to find a directly comparable system for guidance and evaluation.

5.13.4 *Limited Large Scale Real-World Testing*

Another challenge in this project is the difficulty of testing the system with a large number of real users or in a real Hajj environment. Due to limited student resources and the difficulty of accessing real testing scenarios, the system is tested using virtual users through Apache JMeter instead of real participants. The testing is done gradually with 100, 200, 300, 400, and 500 users to evaluate the system response time and performance under different load conditions.

5.13.5 *Defining Risk Weight Values*

Another challenge in this project was selecting suitable weight values for the risk calculation formula. The chosen values (0.6 for crowd density and 0.4 for temperature) were based on logical assumptions about the impact of each factor on risk levels, rather than being taken directly from a specific scientific study related to Hajj environments. Although these values support the system design and risk assessment process.



5.14 Future Work

Planned for GP2

1. Improve the CNN model accuracy using larger and more diverse crowd datasets.
2. Expand scalability and performance testing using Apache JMeter with a larger number of concurrent virtual users beyond the initial 500-user target.

Long-Term Future Work

1. Future improvements may include integrating real-time cameras and environmental sensors to enable live crowd monitoring and real-time temperature analysis instead of relying on simulated or pre-collected data.
2. Future improvements may include integrating wearable devices, such as smart watches, smart bracelets, or smart rings, to collect real-time health and environmental data and improve the accuracy of heat-risk detection.
3. The system may also be extended using Augmented Reality (AR) technologies to visualize zone risk levels and alerts directly on monitoring maps within the dashboard, helping authorities and medical teams identify high-risk areas more efficiently and respond more quickly.
4. Future improvements may include integrating GPS-based location services to automatically identify the pilgrim's current zone instead of relying on manual zone selection, improving the accuracy of location-based risk assessment and alert generation.



5.15 Conclusion

This project addressed the health and safety challenges in crowded, high-temperature environments such as Hajj. A review of existing literature (presented in Chapter 2) revealed that current systems treat crowd monitoring, heat-related health risks, and symptom-based applications separately, leading to a clear research gap: the absence of an integrated system that combines all three factors within a unified framework.

To bridge this gap, the Darp system was designed to integrate three key risk factors: crowd density (estimated via a CNN), environmental temperature (normalized within the realistic Hajj range of 30 – 45°C), and user-reported symptoms (dizziness, fatigue, itching) adjusted by a rule-based engine (+1 for a single report, +2 for three or more).

The system computes a weighted baseline risk score using the formula:

$$0.6 \times \text{Density} + 0.4 \times \text{Normalized Temperature}$$

then classifies the final score into four alert levels (Low, Preventive Warning, High Risk, Critical). Medical teams receive an immediate notification when a zone becomes critical.

All GP-I objectives were successfully achieved, including measurable functional and non-functional requirements, complete system design artifacts (ER, class, DFD, architecture), justified high-level algorithms with pseudocode, and a realistic testing strategy aligned with the non-functional requirements.

The GP-I phase has successfully established a solid foundation for the next phase of the project.



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